

Summary

Qingyuan YAO



Graduate from the AI master's program at Sorbonne University. Experienced in Unity and self-learning Unreal Engine 5. Capable of implementing HCI and AI related theories into simulated environment.

Email	marvinyao@icloud.com
Tel	+330663392132
Portfolio	https://mrzny.github.io/
Github	https://github.com/MRVNY

Education

2021 - 2023	Master of Artificial Intelligence <i>Parcours AgeNts Distribués, Robotique, Recherche Opérationnelle, Interaction, DECision (ANDROIDE)</i> Mention Bien Sorbonne University
2018 - 2021	Bachelor of Computer Science Mention Bien Sorbonne University

Projects

追忆 (Zhuīyì) - In Search of Lost Words

An FPS game that teaches people to write 5 simple Chinese characters.

- Wacom graphic tablet and Nintendo Switch Joycon as input methods (keyboard and mouse are optional)
- Implemented the \$P recognizer for Chinese character recognition
- Implemented Bayesian Knowledge Tracing (BKT) for player modeling and level suggestion

SPY

Enhancement of the serious game SPY that teaches children basic programming notions.

- Added a level editor, a visual novel UI and a new map
- Used Competence-based Knowledge Space Theory (CbKST) to organize levels in a skill tree structure
- Used xAPI to trace user actions

Game Jams

- Developed games in Unity and UE5 in 48 or 72 hours
- Experienced in coming up with games ideas, dissect into small tasks, and learn to implement novel mechanics in a short period of time
- Managed international collaboration in 3 timezones
- Collaborated with other programmers and artists

Skills

- **Programming:** Python, C, C#, C++, Java, JS, LaTeX
- **Game development**
 - **Unity (Experienced):** Targeted for desktop, mobile, web, version control with git, structured C# code, level building, basic materials & particle system, asynchronous & parallel execution, etc.
 - **Unreal Engine 5 (Beginner):** Targeted for desktop, version control with git, code with C++ and Blueprint, terraform and assets to build environment, able to translate game dev experience from Unity
- **Serious Games & Human-Computer Interaction:** Entity-Component System, Design Protocols, User Tracing, User Modeling, Adaptation, etc.
- **Artificial Intelligence:** Deep/Reinforcement Learning, Multi-Agent System, Game Theory, Operational Research, Decision
- **Software and Web Development:** Object-Oriented Programming, Design Patterns, UML, basic frontend & backend with Javascript, HTML, CSS, React
- **Spoken languages:** Chinese (native speaker), English (C2), French (C1/C2), Spanish (A2/B1), Japanese (A2)
- **Media Production:** Lightroom, Final Cut, Logic (limited competence: Procreate, Blender, Photoshop)

Experience

Deep Generative AI Internship Tokyo National Institute of Informatics

March - August 2023

- Created a new dataset of audio-text-motion by extracting motion data from dance videos using existing deep learning models of 3D pose estimation
- Implemented a Transformer VAE baseline model to showcase the validity of adding lyrics as a condition in the process of dance motion generation.

Serious Game Development Internship Team MOCAH of LIP6 at Sorbonne University

June - July 2022

- Solo developer of on the prototype targeted for PC and Android, developed by previous team of students
- Significantly improved stability and optimization by:
 - Fixing collisions and sorting layer issues
 - Using Coroutine, async for asynchronous execution
 - Using basic Jobs System and DOTs for multi-threading
 - Defining better structure with Heritage and Singleton
- Added new implementations into the game such as:
 - A visual novel interface with loadable conversation tree
 - Custom 2D A* pathfinding system
 - Character instant switching system
 - More extensive save&load system
- Communicated with the supervisor on a daily basis and proposed new implementation ideas

Game Design Workshop

2019 - 2022

- Learned the basics of game design and how game designers think
- Made 2 board-game prototypes (cf. portfolio)